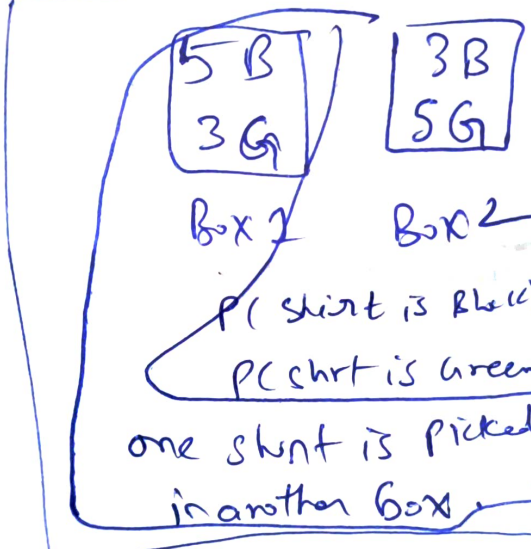
General Aptitude
- Data interpretation & Logical Reasoning
- Quant
- verbal ability & Reading Comprehension

Programming
- D.S.A
- Python
- SQL

DataScience
- Stats
- M.L
- D.L

34 mcq, 2 Subjective, 1 Prog., 1 SQL ques.

Total ques :- 38.
Total sec : 20
Total Score :- 125



5 B
3 G

Umesh = 4 * Reena

$\frac{(TP+FN)}{(all)}$ x $x+6$

$u = 7r$
$m = 6r$

$u+6 = 4(\cdot$ $4(r+6) = 7r$

11)

	A	B	C
A	14	0	0
B	1	15	0
C	0	0	6

find accuracy.

A

Pseudocode

```
int func (int arr[], int n, int k)
{
    int res[] = new int [n+1];
        res[0] = 0;
        res[1] = arr[0];
    for (int i = 2 ; i ≤ n; i++)
            res[i] = res[i-1] + arr[i-1]
    PriorityQueue<Integer> p = new priorityQueue<Integer>();
    for (int i = i; i ≤ n; i++)
    {
        for (int j = 1; j ≤ n; ++j)
        {
            int res2 = res[j] - res[i-1];
            if (p.size() < k)
                p.add(res1);
            else
                if (p.peek() < res1)
                {
                    p.poll(); p.add(res1);
```

```
return p.poll();
```

arr[] = {2, -5, -7, 6, 4, 3, -10}

k = 3, n = 7.

O/P of program

```
import pandas as pd
df = pd.read_csv("sales.csv")
#1. Drop columns where all values are missing
df = df.____(axis=1, how='all')
#2. Remove duplicate rows, keeping the first occurence
df = df.____(keep='____')
#3. Fill missing values in 'revenue' with col. median
df['revenue'] = df['revenue'].____(df['revenu'].median())
#4. Normalise col. names: lowercase + replace spaces with underscores
df.columns = df.columns.____.____('_')
#
```

^ step A ^ step B

upstream speed = 22 km/hr
downstream speed = 34 km/hr
Speed of boat in still water?

Assume simple MLP model with 3 neurons & I/P = 1, 2, 3. weights of (neurons I/P) are 4, 5, 6
Assume activation func. is linear const. value of 3.
What will be O/P? a) 32 c) 96
b) 64 d) 128.

delectable Synonym
- horrible
- platable
- disagreeable
- Tasteless

Tasks:
1. Fill all 4 blanks with correct pandas Methods/arguments
2. for blank 2 – why is keep='first' the safe default? Name one situation where you would use keep=False instead.
3. For blank 4 – why clean col. names this way before modelling?

Write reusable func. quick_eda(df) that prints a structured summary automatically.
Sample dataset func will be tested on.

customer_id	age	city	purchase_amount	subscribed
1001	34	Mumbai	2500.0	Yes
1002	NaN	Delhi	1800.0	No
1003	29	Mumbai	NaN	Yes
1004	45	Chennai	3200.0	Yes
1005	NaN	pelhi	1800.0	No

Your func must print
1. Shape of dataframe – rows & columns
2. Data type of each column.
3. Count & percentage of missing values per column (only show col. that have atleast one missing value)
4. No. of duplicate rows 5. For numeric columns only – mean, median, min, max,

Consider a trial of criminal. If type I error has occured in judgement then which of the following statement is true?

a) guilty person is set free
b) an innocent person is convicted
c) a guilty person is convicted
d) an innocent person is set free.

In which process do discrete events occur in a cont'n but finite interval of time/space in Data science?

a) Poisson distri c) Linear regression
b) Binomial distri d) Lambda expression

What is distinctions b/w popular parameter & sample statistic?

In 1929, Babe Ruth had 522 bats with 205 hits at
Assuming that binomial distribution can be appropriately applied, find expected no. of hits in 529 at bats?

a) 321 b) 186 c) 230 d) 208

Alice & Bob play a series of 5 chess games.
Prob. that Alice wins a game is 2/5.
Bob wins a game is 3/5.
There is no prob. of a draw.
The series will be won by person who wins 3 matches. The series ends at moment when any of 2 wins 3 matches.
determine prob. of Alice winning series?

a) 992/3125 b) 273/625 c) 1021/3125 d) 1081/3125

In model based learning methods, an iterative process takes place on ML models that are built based on various model parameters?

a) mini-batches b) optimized param c) hyper param d) super param

PCA

which are True for PCA?

a) it performs mean normalization
b) it creates an embedding of dataset & attempts to preserve relationships in dataset
c) it uses covariance matrix & eigenvectors for its implementation
d) data std. is not required before implem
e) every principal component is always orthogonal

What is accuracy based on following confusion matrix $C = \begin{bmatrix} 14 & 0 & 0 \\ 1 & 15 & 0 \\ 0 & 0 & 6 \end{bmatrix}$

0.75, 0.97, 0.95, 0.85

which activ. func. led to vanishing gradients?

ReLu, Tanh, LeakyReLu, None of above

Suppose u have to minimize cost func. by changing the parameters. Which of following techniques could be used for this?

a) Exhaustive search b) Random search c) Bayesian optimi
d) All of above

u need to classify from dataset which configuration in output layer?

activ. func	Loss func
Softmax	cross entropy
Sigmoid	cross entropy
Softmax	mean square error
sigmoid	mean squared error